The JACON

(Jason's RACON draft design)

9.3–9.5 GHz
50 ns – 1.2 µs
radar pulse

RX Antenna

Horn Antenna with SMA (testing):
≈$800
8.2GHz-12.4GHz
10dBi~15dBi Gain

9.3–9.5 GHz (200 MHz BW)
−50 dBm (distant radar) to
−10 dBm (close-range radar)

Power Limiter

Aaronia passive "SMA Limiter 20GHz" 250 EUR but
  POut @ 10GHz = 17 dBm
Pasternack PE8035 Limiter 6GHz ~ 18GHz $600 but Obsolete
Marki Microwave HLM-20PSM (DC–20 GHz) EVAL-HLM-20P SMA $354 but limited availability
Pasternack PE80L2006 Limiter 2GHz ~ 18GHz $1518
Fairview Microwave FMLM2013 (active) $6000

If the power limiter allows more than +15 dBm through,
add an in-line attenuator to get down to +15 dBm.
This is bad for sensitivity, but keeps LNA safe for testing.

LNA

Low Noise Amplifier
ZX60-5123LN+ ≈$310
5.5–12.5GHz; +22 dB
low noise 1.4 dB
Input Power Limit +15 dBm
High Output P1dB, +17 dBm
12V DC Operating Voltage

9.3–9.5 GHz (200 MHz BW)
−30 to +10 dBm

Mixer

ZX05-14-S+ (LO level +7 dBm no stock) or
ZX05-14LH-S+ (LO level +10 dBm) (yes, 16 in stock)
RF and LO: 3.7-10 GHz
IF BW: DC to 4 GHz
6.7 dB conversion loss
≈ $70

1.0–1.2 GHz (200 MHz BW)
−37 to +3 dBm

IF amp

Maybe necessary for wide dynamic range
ZX60-2534MA-S+ 500-2500MHz 43 dB gain $113
ZX60-P162LN+ 700-1600MHz 22.5 dB gain $130 (yes)
ZX60-P103LN+ 50-3000MHz 13 dB gain $120
ADL5610-EVALZ ≈ $165 or
PCB: Mini-Circuits GALI- or ERA- series

50Ω Splitter

ZFRSC-42-S+ (IF/audio, DC-4200 MHz,
2-way resistive, SMA): >100 in stock, $78.86
outputs feed independent detectors,
isolation between them doesn't matter much here

RF Limiting Amplifier

Amplifies input to a fixed level
Three cascaded Mini-Circuits gain blocks
(from the IF amp options above)
(for PCB, GALI-52+ / ERA-33SM+), ~15–18 dB each,
last stage(s) deliberately driven into mild saturation

50Ω Splitter

ZFRSC-42-S+ again -- but this is the
delay-line split (direct vs delayed copy),
more isolation-sensitive than the other one.
Resistive = great balance, only ~11dB isolation.
In-stock Wilkinson (ZX10-2-12/-20-S+) = 0 stock,
30+wk lead -- OK for now, revisit if needed.

Extra delay Δt≈0.2ns (3-4 cm) calculated such that
middle of range (1.1 GHz IF for 9.4 GHz RF)
gives 90 degrees of phase shift and is on
linear region of the resulting cos(2πfΔt).

Phase detector

Passive Double Balanced Mixer
ZX05-C24LH-S+ Level 10, 300 - 2400 MHz $55
ZX05-30W-S+ Level 7, 300 - 4000 MHz $55 (yes)
Analog Devices ADL5380 is perfect but too slow
ADL5380-EVALZ is $163; chip is $65

Sample and Hold

ADG1211 Analog Switch $7 (no eval board?)
TODO: check turn on and off times
100 pF C0G Capacitor to Hold
ADA4622-1 dual op-amp $6
to shift and scale

±5V or ±12V

Bipolar, dual rail power supply
for Op Amps and "slow" (non-RF)
analog signals

50Ω Splitter

ZX10-2-1252-S+
2 Ways DC Pass Power Splitter, 1800 - 12500 MHz
Low insertion loss, 0.7 dB $92

LO Buffer Amp

Mini-Circuits ZX60-183-S+ (no stock)
6–18 GHz, SMA connectorized
24 dB gain, +18 dBm P1dB
Mini-Circuits ZX60-14012L-S+
0.3 MHz to 14 GHz, +12dB, $340

PLL Local Oscillator

ADF5355 module
Run it at 8.3 GHz
Output power gets split and then needs
to match mixer's expected LO power
≈ $250

DAC

to set comparator level to trigger
Analog Devices nanoDAC series

Comparator

high when ship's radar pings
EVAL-ADCMP600B $82
ADCMP600 chip $6

FPGA

control, Morse sequencing, hold off
PYNQ-Z2 ≈ $100
or ECP5 for more open source

Mixer

(same as for RX)

Keying switch

EVAL-ADG901EBZ ≈ $100
ADG901 chip ≈ $5

Image Reject Filter

Pass 9.3-9.5 GHz
filter out 7.1-7.3 GHz image
MCL VHF-8400+ high-pass, 8.4GHz cutoff
9.3-9.5GHz: 1.4-1.6dB IL; ~24-28dB rejection
at 7.1-7.3GHz (interpolated from datasheet)
216 in stock at minicircuits.com, $33.81

Power Amp

ZVE-143-S+ ($1,980, >10 stock, 8-14 GHz),
ZVE-143X-S+ (no heatsink)
ZVA-183G-S+ ($2,595, 0.5-18 GHz, ~1/2W),
ZVA-183A-S+ ($2,105, 0.7-18 GHz, +24 dBm P1dB)
ADPA9002 ($1,100, 2W, Digi-Key stock

TX Antenna

a copy of RX antenna for testing...
or add a circulator and/or
antenna switch to share antennas

9.3–9.5 GHz
1W or so
Pulsed Morse

VCO

Voltage Controlled Oscillator
ROS-1500+ (chip; end of life)
Mini-Circuit modules with connectors
are all end-of-life and don't go high in frequency
Crystek CVCO55BE-0800-1600 VCO chip
  247 in stock at Digi-Key, $24.77
Crystek CEVAL-055 matching SMA breakout board
  12 in stock, $60.96
Together ≈$86
Tuning curve shows 1.0–1.2 GHz only needs ~6–9.5V
on the tune pin; separate Vcc pin needs ~11.5V
CEVAL-055 ships BLANK -- solder VCO on yourself